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Distributionally Robust Newsvendor Under Stochastic Dominance with a Feature-Based Application

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Abstract. ***Problem definition*:** In this paper, we study the newsvendor problem under some distributional ambiguity sets and explore their relations. Additionally, we explore the benefits of implementing this robust solution in the feature-based newsvendor problem. ***Methodology and results*:** We propose a new type of discrepancy-based ambiguity set, the JW ambiguity set, and analyze it within the framework of first-order stochastic dominance. We show that the distributionally robust optimization (DRO) problem with this ambiguity set admits a closed-form solution for the newsvendor loss. This result also implies that the newsvendor problem under the well-known infinity-Wasserstein ambiguity set and Lévy ball ambiguity set admit closed-form inventory levels as a by-product. In the application of feature-based newsvendor, we adopt general kernel methods to estimate the conditional demand distribution and apply our proposed DRO solutions to account for the estimation error. ***Managerial implications*:** The closed-form solutions enable an efficient computation of optimal inventory levels. In addition, we explore the property of optimal robust inventory levels with respect to the nonrobust version via concepts of perceived critical ratio and mean repulsion. The results of numerical experiments and the case study indicate that the proposed model outperforms other state-of-the-art approaches, particularly in environments where demand is influenced by covariates and difficult to estimate.

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Keywords: distributionally robust optimization • newsvendor problem • demand covariates • kernel estimation • Wasserstein distance • asymptotic convergence • stochastic dominance • Kullback–Leibler divergence

1. Introduction

The newsvendor problem is perhaps the most fundamental problem in operations research and operations management, and it has numerous applications in inventory management, revenue management, and the healthcare industry. The newsvendor solution carefully balances the overage cost of ordering too much and the opportunity cost of ordering too little. If the demand distribution is explicitly known, the optimal inventory level equals the quantile of the so-called *critical ratio*, which is the proportion of the underage cost to the sum of the overage and the underage costs. In reality, the demand distribution is rarely known. Instead, historical demand samples might be available. Using these samples, one might fit some parametric classes of demand distribution or directly use the empirical distribution and apply the newsvendor solution. However, these data samples might not be representative of new demand scenarios. It is well-known that the newsvendor solution may perform poorly if the demand distribution is misspecified. For this reason, distributionally robust optimization (DRO) has been applied to the newsvendor problem ever since the seminal study in Scarf et al. (1957). The goal of DRO is to find an ordering decision that performs well under the worst-case demand distribution. By carefully designing the ambiguity set, data-driven DRO can play the role of regularization (see, e.g., Shafieezadeh-Abadeh et al. 2019 and Duchi et al. 2021) and thus better control the bias-variance trade-off.

In this paper, we study the distributionally robust newsvendor problem under some ambiguity sets that

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were not studied in literature. We start with a new distance-based ambiguity set, which we refer to as the JW ambiguity set. The JW ambiguity set includes all distributions that are not far from the reference distribution based on the distance measure proposed by Jager and Wellner (2007), which we refer to as the JW discrepancy. The JW discrepancy is based on the ϕ-divergence for the probability that the random variable is less or greater than a specified threshold. Unlike the well-known ϕ-divergence ambiguity set proposed in Ben-Tal et al. (2013), the JW ambiguity set takes into account both demand scenarios and choice probability, and it contains both discrete and continuous distributions, even if the reference distribution is discrete. More detailed discussions on the relationship between these ambiguity sets are presented in Section 3.1.

We find that the JW ambiguity set belongs to a more general ambiguity set that is referred to as the first-order stochastic dominance ambiguity set (the FSD ambiguity set), which includes all of the distributions that lie between two distributions in the first-order stochastic dominance sense. Besides the JW-ambiguity set, we show that FSD ambiguity also includes the well-known ∞-Wasserstein ball and Lévy ball as special cases.

Then, we investigate the optimal DRO inventory decision under the FSD ambiguity set in Section 4. Interestingly, we find that the distributionally robust newsvendor problem under the general FSD ambiguity set admits a simple, closed-form optimal inventory solution. Therefore, computing the optimal inventory decision in the DRO setting is almost as easy as computing the optimal inventory level for its stochastic counterpart. Similar to distributional ambiguity sets that are based on the reference distribution, the robustness level can be adjusted based on the size of the data samples and the risk attitude of the decision maker. To obtain more insights into the closed-form inventory level, we propose the concept of median repulsion and explore the conditions for the reference distributions that satisfy this property. We test the performance of the proposed ambiguity set via a set of numerical experiments in Section 5.

In the big data era, a challenge (or opportunity) faced by the real-world newsvendor problem is that demand typically has various covariates (also called features or attributes), such as price, purchase date, and region. In other words, historical demand is usually collected at different price points, on different days, or through different sales channels. Therefore, it is essential to consider this covariate information when predicting demand. The problem of determining the optimal inventory level based on covariates is usually called the *feature-based newsvendor problem*. There are two typical approaches to this problem. The first approach assumes a parametric relationship between the true demand and the covariates and directly predicts the optimal order quantity based on these covariates. For example, quantile regression (e.g., as in Koenker and Hallock 2001) assumes that demand linearly depends on the covariates and directly predicts the optimal inventory decision based on the critical ratio. However, in real-world applications, the linearity assumption can be restrictive. The second approach estimates the demand distribution based on the covariates and then applies the newsvendor solution. For instance, Ban and Rudin (2019) proposed using kernel methods to estimate the distribution and obtain the optimal order quantity. Although this approach can handle a more versatile relationship between demand and covariates, the estimated demand distribution may be inaccurate, and a decision based purely on this demand could lead to significant loss of profit.

In Section 6, we apply the estimate-then-optimize approach to address the feature-based newsvendor problem in the presence of both demand covariates and demand uncertainty. Our approach first estimates the conditional demand distribution based on the covariates using kernel methods. We then apply the Distributionally Robust Optimization framework with the JW ambiguity set to mitigate errors in estimating the conditional distribution. Thanks to the closed-form solution developed for the JW ambiguity set, the entire procedure can be efficiently executed with efforts comparable to its nonrobust version. In addition, the performance of this approach is theoretically analyzed in Online Appendix E. We implement our proposed approach on a publicly available e-commerce data set in the context of a new product launch, testing its practical performance and computational efficiency against other state-of-the-art approaches.

The remainder of this paper is structured as follows. Section 2 reviews the related literature. Section 3 introduces the problem settings and the ambiguity sets studied in this paper. Section 4 presents the closed-form solution to our distributionally robust newsvendor problem and certain properties of the solution. Section 5 presents a numerical study that examines the performance of our proposed DRO approach. In Section 6, we apply our DRO approach to a feature-based newsvendor problem; we first describe the problem setting and decision approach, then conduct a case study with a real-world data set to evaluate the performance of our proposed approach. Section 7 concludes the paper.

1.1. Notation

In this paper, we use boldface lowercase letters for vectors or arrays (e.g., $\boldsymbol{d}$). Each random variable (vector) is denoted with a tilde (e.g., $\tilde{d}$ or $\tilde{\boldsymbol{d}}$). As this paper focuses mainly on the cumulative distribution function (CDF), we use this function (e.g., F) to indicate a distribution

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and $\mathbb{E}_F[f(\tilde{d})]$ to represent the expectation of a function $f(\tilde{d})$, with $\tilde{d}$'s CDF being F. $\mathcal{P}(\Omega)$ is the set of all distributions with sample space Ω. The order relationship between the vectors implies an element-wise relationship; for example, $\boldsymbol{a} \leq \boldsymbol{b}$ means $a_i \leq b_i$ for all possible i. For two distributions, F_1 and F_2, we use $F_1 \preccurlyeq F_2$ to denote that the F_2 first-order stochastically dominates F_1; that is, the CDF of F_2 is uniformly larger than or equal to F_1. In particular, we use $(x)_+$ to indicate $\max\{x, 0\}$.

2. Literature Review

In this section, we review the literature related to the distributionally robust newsvendor problem and feature-based newsvendor problem, as well as the estimation method that we used in our feature-based application.

2.1. DRO for Newsvendor Problem

Ever since its introduction, DRO has been widely applied to the newsvendor problem. The earliest DRO framework was proposed in Scarf et al. (1957) targeting the newsvendor problem. It offered an inventory solution that is robust to any demand distribution with a given mean and variance. Most DRO models can be roughly divided into two categories, according to their ambiguity set. The first type of ambiguity set contains all distributions with common statistical properties. Scarf et al. (1957) considered all distributions having the same mean and variance. Natarajan et al. (2018) considered all distributions with a given mean, variance, and semivariance. Ardestani-Jaafari and Delage (2016) studied a newsvendor problem with a given mean and first-order partial moments (with the conditional expectation of an upside or downside tail with respect to a threshold). Ambiguity sets in Perakis and Roels (2008) incorporate demand information such as mean, median, variance, unimodality, and symmetry. Das et al. (2021) extended Scarf's model to a case in which the ambiguity set is determined by the first and α-th moment. Govindarajan et al. (2021) studied the inventory polling problem with the ambiguity set determined by the mean and covariance. Besbes and Mouchtaki (2023) investigated the worst-case relative regret of data-driven newsvendor policies, and Besbes et al. (2023) extended this framework into a contextual setting.

The second type of distributional ambiguity set contains all of the distributions close to a reference distribution (usually the estimated demand distribution) under some discrepancy measures. Different discrepancy measures, such as ϕ-divergence (e.g., Ben-Tal et al. 2013, Hanasusanto and Kuhn 2013, Klabjan et al. 2013, Wang et al. 2016, and Rahimian et al. 2019) and the Wasserstein distance (e.g., Esfahani and Kuhn 2018 and Gao and Kleywegt 2023), have been adopted. Bertsimas et al. (2018) proposed a framework to link the DRO approach with the goodness-of-fit test. The discrepancy measures in their ambiguity set are derived from the hypotheses test statistics. As mentioned above, our proposed JW ambiguity set is based on the discrepancy measure first proposed in Jager and Wellner (2007).

Closed-form inventory solutions for newsvendor problem are available for some special ambiguity sets. Esfahani and Kuhn (2018) showed that the DRO solution with the 1-Wasserstein metric and unbounded sample space is the same as the sample average approximation (SAA) solution. Lee et al. (2021) developed a closed-form solution for the p-Wasserstein DRO model with $p > 1$, under the assumption that the critical ratio is larger than half and the demand is nonnegative. Proposition 7 of Bertsimas et al. (2018) considered the Kolmogorov-Smirnov (KS) distance ambiguity set based on the empirical distribution and obtained a closed-form solution for newsvendor-type problems. This proposition can be viewed as a special case of Theorem 1 in this paper, because the JW discrepancy measure includes the KS distance as a special case. In Subsection 5.2, we use numerical experiments to show that the additional flexibility of our proposed JW ambiguity set helps produce more reasonable inventory decisions.

2.2. The Feature-Based Newsvendor Problem

There is a recent trend in studying the newsvendor problem with demand features. This problem involves a demand distribution that depends on various observed features before decision making. One popular approach to solving this problem is to formulate it as a quantile regression problem (e.g., Koenker and Hallock 2001 and Ferraty and Vieu 2006), which is possible because the optimal inventory decision is a quantile of the actual distribution conditioning on the demand features. See and Sim (2010) assumed that historical demand is a time series and that true demand depends on the current period and certain random features. Beutel and Minner (2012), Sachs (2015), and Ban and Rudin (2019) used linear programming to find the optimal linear inventory function that minimizes empirical loss. Qi et al. (2022) proposed a regress-then-robustify framework to tackle the feature-based newsvendor problem for the linear relation function. They assumed that these samples originated from a fixed design, rather than being independent and identically distributed, which introduced additional challenges.

Another popular approach is the estimate-then-optimize approach, in which decision makers first estimate the conditional distribution of demand and then decide the optimal order quantity based on the estimated distribution. In this paper, we adopt this approach and utilize a general kernel estimation to estimate the demand distribution. The kernel estimation was first introduced by Nadaraya (1964) and Watson (1964). It has been

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widely used in regression analysis and distribution estimation and has inspired other nonparametric estimation methods (e.g., Fan 1993, Seifert and Gasser 1996, and Hall et al. 2012) and has been applied to conditional stochastic optimization problems (e.g., Hanasusanto and Kuhn 2013, Bertsimas and Kallus 2020, and Srivastava et al. 2021). The kernel estimation in this paper is based on semimetric (a metric, but does not require the identities of indiscernibles), which is an extension of the Nadaraya-Watson estimator. With the proper choice of semimetric and kernel function, various machine learning methods, such as classification and regression tree (CART) (e.g., Breiman et al. 1984) and random forest (RF) (e.g., Breiman 2001), can be applied to estimate the demand function (see, e.g., Scornet 2016). Similar nonparametric weighted neighbor estimators, such as k-nearest neighbors, Dirichlet kernel estimation, Nadaraya-Watson (NW) kernel estimation, CART, and RF, have been adopted to estimate the demand distribution (e.g., Hannah et al. 2010, Ban and Rudin 2019, and Lin et al. 2022).

Similar to this paper, some other papers adopt the DRO approach in the optimization stage. Nguyen et al. (2020) and Zhang et al. (2024) considered the Wasserstein ambiguity sets, which include joint distributions of the response and predictors that are not far from the empirical distribution. In contrast, we consider only the distributional ambiguity in the response variable in our ambiguity set. Wang et al. (2021) proposed a DRO approach similar to ours. They also used kernel methods to estimate the conditional demand distribution, but they adopted the Wasserstein ambiguity set. They provided a tractable convex optimization formulation for the newsvendor and the portfolio optimization problems, with extensions to more general functions. Our paper differs from Wang et al. (2021) in the following two aspects. First, we derive a closed-form inventory solution to the DRO problem with the FSD ambiguity set, which includes the JW ambiguity set, the ∞-Wasserstein ambiguity set, and the Lévy ball ambiguity set as special cases. Second, our assumption on the relationship between demands and features is less restrictive. More details on this distinction can be found in Online Appendix E.

3. Problem Formulation

In this paper, we study the classic newsvendor problem, in which a seller decides how many products to order (or produce) based on past demand information to balance the underage and overage costs. Let $\tilde{d} \in [d_l, d_u]$ be the random demand. Given the order quantity decision q and the random demand $\tilde{d}$, the newsvendor loss function is $l_0(q,\tilde{d}) = \max\{b(\tilde{d}-q), h(q-\tilde{d})\}$, where b and h are the shortage and holding costs, respectively. The seller makes an ordering decision based on the historical information to minimize the expectation of the newsvendor loss $l_0(q,\tilde{d})$ over the random demand $\tilde{d}$. As the exact demand distribution is unknown, we apply the DRO framework, assuming that the real demand distribution lies in an ambiguity set that depends on the demand covariates. In other words, the problem we want to solve is

$$\min_{q} \sup_{F\in\mathcal{F}} \quad \mathbb{E}_F[l_0(q,\tilde{d})], \tag{1}$$

where $\mathcal{F}$ is the distributional ambiguity set.

Note that, in practice, demand usually depends on several observable features, such as weather, location, season, and economic indicators. In this case, the ambiguity set should depend on the observed feature. We will discuss the corresponding settings in Section 6.

3.1. JW Ambiguity Set

The distribution of demand can usually be estimated from historical data through various approaches. Therefore, it is natural to consider the distributional ambiguity set that contains all distributions not too far from the estimated demand distribution, $\hat{F}(d)$. In this paper, we mainly study the distance measure proposed in Jager and Wellner (2007). The distance measures the discrepancy between two distributions based on the ϕ-divergence, which is defined as follows.

Definition 1. The ϕ-divergence between two vectors $\boldsymbol{u} = (u_1, \dots u_n) \geq \mathbf{0}$ and $\boldsymbol{v} = (v_1, \dots v_n) \geq \mathbf{0}$ is defined as $I_\phi(\boldsymbol{u},\boldsymbol{v}) := \sum_{i=1}^n v_i \phi\left(\frac{u_i}{v_i}\right)$, where $\phi(t)$ is a continuous convex function with domain $t \geq 0$ and $\phi(1) = 0$. Furthermore, $\phi(t)$ strictly decreases in $[0,1]$ and strictly increases in $[1,\infty)$. By convention, $0\phi(a/0)$ is defined as $a \lim_{t\to\infty}\phi(t)/t$ and $0\phi(0/0) := 0$.

Then, the discrepancy measure between the two distribution functions proposed in Jager and Wellner (2007), referred to as the *JW discrepancy*, is defined as follows.

Definition 2. Given two distributions with CDF $\hat{F}(d)$ and $F(d)$, the JW discrepancy measure $S_\phi(\hat{F},F)$ is defined as $S_\phi(\hat{F},F) := \sup_d \kappa_\phi(\hat{F}(d),F(d))$, where $\kappa_\phi(u,v) := I_\phi((u,1-u),(v,1-v))$ for all $u,v \in [0,1]$.

The basic idea of this discrepancy measure is to divide the random demand $\tilde{d}$ into two parts, $\tilde{d} \leq d$ and $\tilde{d} > d$. The indicator function $\mathbf{1}_{\tilde{d}\leq d}$ represents the new Bernoulli random variable generated by the division. Then, $S_\phi(\cdot,\cdot)$ is the maximum ϕ-divergence between the two Bernoulli random variables among all thresholds d.

In Table 5 in Online Appendix A, we give some examples of the JW discrepancy measure associated with the Cressie and Read ϕ-divergence class proposed in Cressie and Read (1984), which is defined as $\phi_{cr}^\theta = \frac{1-\theta+\theta t-t^\theta}{\theta(1-\theta)}$, where $I_\phi(\boldsymbol{u},\boldsymbol{v}) = (1-\sum_{i=1}^n u_i^\theta v_i^{1-\theta})/(\theta(1-\theta))$.

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Based on the JW discrepancy measure, we now define the *JW ambiguity set* $\mathcal{F}_\phi(\hat{F},\rho)$.

Definition 3. Assume $\hat{F}(d)$ is the reference distribution function. Then, the *JW ambiguity set* with sample space Ω is defined as

$$\mathcal{F}_\phi(\hat{F},\rho) = \{F \in \mathcal{P}(\Omega) : S_\phi(\hat{F},F) \le \rho\}. \tag{2}$$

The JW ambiguity set is closely related to the well-known ϕ-divergence ambiguity set proposed in Ben-Tal et al. (2013). Compared with the ϕ-divergence ambiguity set, the JW ambiguity set has two salient features. First, the JW ambiguity set contains discrete and continuous distributions, even if the reference distribution is discrete. However, the ϕ-divergence ambiguity set includes only distributions with exactly the same support as the reference distribution. This means that the ground-truth distribution, if it exists, is more likely to be outside the ambiguity set. More importantly, when measuring the distance of two distributions, the ϕ-divergence measure only considers the difference in probability measure, not the demand support. More specifically, if the probability is moved from one demand value to another, the ϕ-divergence measure does not account for the distance between the two demand values. This property is inconsistent with most other discrepancy measures, such as the Wasserstein distance, and may have unusual implications in the corresponding newsvendor solution, as shown in Section 4.1. Nevertheless, the JW discrepancy measure takes both the order information between demand scenarios and the corresponding occurring probability into account, and, thus, it is more consistent with the Wasserstein distance. The following example displays the relations between these discrepancy measures for the case of discrete distributions.

Example 1. Consider distributions with three support points $(d_1,d_2,d_3) = (1,2,4)$. The reference distribution has the probability vector $\boldsymbol{q} = (1/3,1/3,1/3)$. By moving 5/24 density from two support points to the other support point, we obtain three probability vectors: $\boldsymbol{p}_1 = (3/4,1/8,1/8)$, $\boldsymbol{p}_2 = (1/8,3/4,1/8)$, and $\boldsymbol{p}_3 = (1/8, 1/8,3/4)$. The discrepancy between $\boldsymbol{q}$ and $\boldsymbol{p}_i$ under KL-divergence I_{KL} and the JW discrepancy measure based on KL-divergence S_{KL}, ∞-Wasserstein distance W_∞, and 1-Wasserstein distance W_1 are listed as follows.

As shown in Table 1, under KL-divergence, the three cases are the same because the ϕ-divergence measure only considers the change in the probability mass. Under the JW discrepancy measure and the ∞-Wasserstein distance, $\boldsymbol{q}$ is closer to $\boldsymbol{p}_2$ than $\boldsymbol{p}_1$ and $\boldsymbol{p}_3$, but their insights are slightly different. The JW ambiguity set considers the order information of support points, so moving a mass from d_1 or d_3 to d_2 will be less costly than moving it from d_1 to d_3. In contrast, the ∞-Wasserstein distance only considers the maximal distance of the mass movement. Moving from $\boldsymbol{q}$ to $\boldsymbol{p}_1$ (or $\boldsymbol{p}_3$) requires moving a mass from d_1 to d_3 (or vice versa), and, thus, the distance is three. But moving from $\boldsymbol{q}$ to $\boldsymbol{p}_2$ only requires moving the mass from d_2 to d_1 and d_3, so the distance is two. Under the 1-Wasserstein distance measure, which considers the cost of moving probability mass, $\boldsymbol{q}$ is closest to $\boldsymbol{p}_2$, followed by $\boldsymbol{p}_1$ and then $\boldsymbol{p}_3$. Overall, compared to the KL-divergence, the JW discrepancy measure is more consistent with the Wasserstein distance.

Table 1. Discrepancy Between $\boldsymbol{p}_i$ and $\boldsymbol{q}$ Under Different Measures

Discrepancy measure	I_{KL}	S_{KL}	W_∞	W_1
$(\boldsymbol{q},\boldsymbol{p}_1)$	0.3836	0.3836	3	0.8333
$(\boldsymbol{q},\boldsymbol{p}_2)$	0.3836	0.1456	2	0.625
$(\boldsymbol{q},\boldsymbol{p}_3)$	0.3836	0.3836	3	1.0416

3.2. A More General Ambiguity Set

In this subsection, we show that the JW ambiguity set proposed in the previous section is a special case of a more general ambiguity set, which we refer to as the *first-order stochastic dominance ambiguity set*. This ambiguity set contains all possible distributions between a lower-bound distribution F^l and an upper-bound distribution F^u in the sense of first-order stochastic dominance, which is defined as follows.

Definition 4 (First-Order Stochastic Dominance). For any two distributions with CDFs F and G, G first-order stochastically dominates F (denoted as $F \preccurlyeq G$) if and only if $G(x) \le F(x)$, $\forall x$.

Intuitively, if $F \preccurlyeq G$, for any threshold x, the probability of the random variable being larger than x under G is greater than that under F.

Definition 5. Assume that F^l and F^u are two distributions and that F^u first-order stochastically dominates F^l. Then, the first-order stochastic dominance ambiguity set defined by these two distributions is

$$\mathcal{F}_{FSD}(F^l,F^u) = \{F : F^u \preccurlyeq F \preccurlyeq F^l\}. \tag{3}$$

In other words, all of the CDFs in this ambiguity set take values between F^l and F^u. Next, we show that the JW ambiguity set is a special case of the FSD ambiguity set under certain mild conditions.

Proposition 1. *For any $z \in [0,1]$, let $\kappa^{-l}_{\phi,\rho}(z) = \min\{x : x \in [0,z], \kappa_\phi(z,x) \le \rho\}$ and $\kappa^{-u}_{\phi,\rho}(z) = \max\{x : x \in [z,1], \kappa_\phi(z,x) \le \rho\}$. For a JW ambiguity set $\mathcal{F}_\phi(\hat{F},\rho)$, if $F^l(d) = \kappa^{-l}_{\phi,\rho}(\hat{F}(d))$ and $F^u(d) = \kappa^{-u}_{\phi,\rho}(\hat{F}(d))$ are valid CDFs, then $\mathcal{F}_\phi(\hat{F},\rho) = \mathcal{F}_{FSD}(F^l,F^u)$.*

Proposition 1 provides a way to construct the FSD ambiguity set $\mathcal{F}_{FSD}(F^l,F^u)$ from a given JW ambiguity

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Table 2. $\kappa^{-l*}_{\phi,\rho}$ and $\kappa^{-u*}_{\phi,\rho}$ Under Some ϕ-Divergence with Parameter ρ and Reference Distribution $\hat{F}$

Divergence	$\phi(t)$	$\kappa^{-l}_{\phi,\rho}(\hat{F})$	$\kappa^{-u}_{\phi,\rho}(\hat{F})$
χ^2-distance	$\frac{(t-1)^2}{t}$	$\hat{F}-\sqrt{\rho\hat{F}(1-\hat{F})}$	$\hat{F}+\sqrt{\rho\hat{F}(1-\hat{F})}$
Modified χ^2-distance	$(t-1)^2$	$\frac{2\hat{F}+\rho-\sqrt{\rho^2+4\rho\hat{F}(1-\hat{F})}}{2+2\rho}$	$\frac{2\hat{F}+\rho+\sqrt{\rho^2+4\rho\hat{F}(1-\hat{F})}}{2+2\rho}$
Hellinger distance	$(\sqrt{t}-1)^2$	$\rho(1-2\hat{F})+\rho^2(\hat{F}/2-1/4)+\hat{F}$ $-1/2\sqrt{\rho(\rho-2)^2(\rho-4)\hat{F}(\hat{F}-1)}$	$\rho(1-2\hat{F})+\rho^2(\hat{F}/2-1/4)+\hat{F}$ $+1/2\sqrt{\rho(\rho-2)^2(\rho-4)\hat{F}(\hat{F}-1)}$
χ-divergence of order $\theta>1$	$\lvert t-1\rvert^\theta$	$\hat{F}-\left(\frac{\rho\hat{F}^{\theta-1}(1-\hat{F})^{\theta-1}}{\hat{F}^{\theta-1}+(1-\hat{F})^{\theta-1}}\right)^{\frac{1}{\theta}}$	$\hat{F}+\left(\frac{\rho\hat{F}^{\theta-1}(1-\hat{F})^{\theta-1}}{\hat{F}^{\theta-1}+(1-\hat{F})^{\theta-1}}\right)^{\frac{1}{\theta}}$
Variation distance	$\lvert t-1\rvert$	$\hat{F}-\rho$	$\hat{F}+\rho$

Notes. *Because of the limitation of space, we omit the max$\{\ldots,0\}$ and min$\{\ldots,1\}$ here. The actual value of $\kappa^{-l}_{\phi,\rho}$ or $\kappa^{-u}_{\phi,\rho}$ is zero or one if the listed formulation is less or greater than zero or one.

set. Some examples of $\kappa^{-l}_{\phi,\rho}$ and $\kappa^{-u}_{\phi,\rho}$ under different ϕ-divergences are summarized in Table 2.

Note that if the sample space is bounded, it is not difficult to find that F^l, F^u are always valid CDFs. If the sample space is unbounded, F^l, F^u remain valid CDFs for some ϕ-divergence measures—for example, χ^2-distance.

Interestingly, in addition to the JW ambiguity set, the well-known ∞-Wasserstein ball and Lévy ball are also subsumed by the FSD ambiguity set. The definitions of ∞-Wasserstein distance and Lévy distance are presented as follows.

Definition 6 (∞-Wasserstein Distance). Assume that $\tilde{\mu}$ and $\tilde{\nu}$ are random variables on $\mathbb{R}$. The ∞-Wasserstein distance between these two probability measures is defined as

$$W_\infty(\tilde{\mu},\tilde{\nu}) := \inf\{\delta : \exists\gamma\in\Gamma(\tilde{\mu},\tilde{\nu}) \text{ s.t. } \mathbb{P}_\gamma[|\tilde{\mu}-\tilde{\nu}|\le\delta]=1\},$$

where $\Gamma(\tilde{\mu},\tilde{\nu})$ denotes the collection of all joint measures on $\mathbb{R}^2$ with marginals $\tilde{\mu}$ and $\tilde{\nu}$.

Definition 7 (Lévy Distance). For any two distributions on $\mathbb{R}$ with CDFs F and G, respectively, we denote their Lévy distance by

$$D_L(F,G) = \inf\{\varepsilon>0 : F(x-\varepsilon)-\varepsilon \le G(x) \le F(x+\varepsilon)+\varepsilon,\ \forall x\in\mathbb{R}\}.$$

The ∞-Wasserstein ball $\mathcal{F}_{W_\infty}(\hat{F},\rho)$ (or Lévy ball $\mathcal{F}_L(\hat{F},\rho)$ on a bounded sample space $[d_l,d_u]$) with radius ρ around a distribution $\hat{F}$ contains all distributions whose ∞-Wasserstein distance (or Lévy distance) to $\hat{F}$ is less than or equal to ρ—that is,

$$\mathcal{F}_{W_\infty}(\hat{F},\rho) = \{F : W_\infty(\hat{F},F)\le\rho\};$$
$$\mathcal{F}_L(\hat{F},\rho) = \{F : D_\infty(\hat{F},F)\le\rho\}. \tag{4}$$

The following propositions show that these two ambiguity sets also belong to the FSD ambiguity set.

Proposition 2. *Let F be the CDF of a univariate random variable. Then, the ∞-Wasserstein ball $\mathcal{F}_{W_\infty}(F,\rho)$ is an FSD ambiguity set $\mathcal{F}_{FSD}(F^l,F^u)$, where $F^l(d)=F(d-\rho)$ and $F^u(d)=F(d+\rho)$.*

Proposition 3 (Brustle et al. 2020). *Let $\hat{F}$ be the CDF of a random variable on $[d_l,d_u]$. Define $\underline{F}$ and $\overline{F}$ to be the CDFs of one-point distributions on d_l and d_u, respectively. Then, the Lévy ball $\mathcal{F}_L(F,\rho)$ on sample space $[d_l,d_u]$ is an FSD ambiguity set $\mathcal{F}_{FSD}(F^l,F^u)$, where $F^l(d)=\max\{F(d-\rho)-\rho,\underline{F}(d)\}$ and $F^u(d)=\min\{F(d+\rho)+\rho,\overline{F}(d)\}$.*

Therefore, the FSD ambiguity set subsumes many distributionally ambiguity sets as special cases. In the following section, we show that the univariate distributionally robust newsvendor problem under the FSD ambiguity set admits a closed-form solution.

4. The Closed-Form Optimal Inventory Decision

In this section, we derive the closed-form solution for the newsvendor problem under the FSD ambiguity set.

Theorem 1. *Let $F^l(d)$ and $F^u(d)$ be two univariate distributions with $F^u \preccurlyeq F^l$. Suppose $F^l(d)$ and $F^u(d)$ have the same bounded sample space, or $\mathbb{E}_{F^u}[|\tilde{d}|],\mathbb{E}_{F^l}[|\tilde{d}|]<\infty$. Let $q^l=\min\{d : F^l(d)\ge\frac{b}{b+h}\}$, $q^u=\min\{d : F^u(d)\ge\frac{b}{b+h}\}$. Then, the optimal solution to the DRO Newsvendor Problem (1) with ambiguity set $\mathcal{F}_{FSD}(F^l,F^u)$ is $q^r=\frac{bq^l+hq^u}{b+h}$. In addition, the worst-case distribution of q^r is*

$$F^0(d) = \begin{cases} F^u(d), & d<q^u,\\ \dfrac{b}{b+h}, & q^u\le d<q^l,\\ F^l(d), & q^l\le d.\end{cases}$$

Theorem 1 shows that if the uncertainty set is bounded by two distributions in the sense of stochastic first-order dominance, the optimal inventory solution has a closed-form solution. The closed-form solution is illustrated in Figure 1. The reference distribution (in black) lies between F^u and F^l. q^u and q^l correspond to the $\frac{b}{b+h}$ quantiles of F^u and F^l, respectively. Then, the optimal inventory decision is a convex combination of q^u and q^l,

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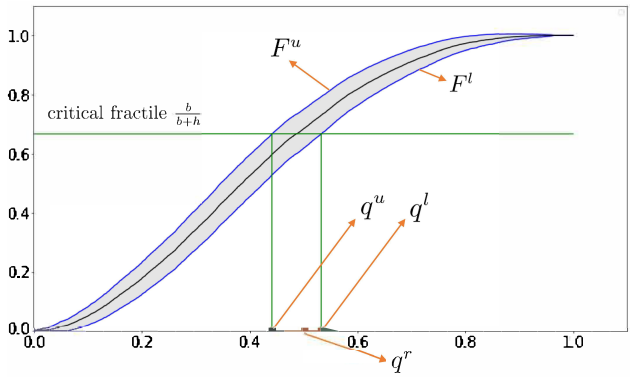
Figure 1. (Color online) The Closed-Form Optimal Solution



and the weight is related to the critical ratio $\frac{b}{b+h}$. Let q^s be the optimal solution to the stochastic inventory problem under the reference distribution; then, q^s is the $\frac{b}{b+h}$ quantile of the reference distribution. Therefore, q^s could be larger or smaller than q^r, depending on the critical ratio and the shape of the region formed by F^u and F^l. If the critical ratio $\frac{b}{b+h}$ is small, the overage cost h is high, so it is optimal to order less than the median of the demand distribution for the stochastic newsvendor problem. In the distributionally robust setting, as the worst-case distribution has a lower quantile, it is optimal to order less than the $\frac{b}{b+h}$ quantile of the reference distribution. In other words, the distributionally robust inventory decision tends to result in an even smaller order quantity to avoid the potentially high overage cost when the overage cost is high. Following the same logic, the optimal DRO inventory level tends to be higher than the inventory level under the reference distribution when the underage cost is high.

By directly applying Theorem 1, we obtain the closed-form expression for the optimal distributionally robust inventory decision under the JW ambiguity set when the reference distribution is discrete.

Corollary 1. *Assume that the reference distribution $\hat{F}$ is a discrete distribution supported on $\{d_0 = d_l, d_1, \ldots, d_N, d_{N+1} = d_u\}$ and the ambiguity set is $\mathcal{F}_\phi(\hat{F}, \rho)$. Let $\xi_i = \kappa_\phi(\hat{F}(d_i), \frac{b}{b+h})$ for $i = 0, 1, \ldots, N$. If $\xi_i > \rho$ for all i, the optimal solution to the DRO Newsvendor Problem (1) is $q^r = \min\{d : \hat{F}(d) \geq \frac{b}{b+h}\}$. Otherwise, the optimal solution is $q^r = \frac{bd_{i_2} + hd_{i_1}}{b+h}$, where $i_1 = \min\{i : \xi_i \leq \rho, 0 \leq i \leq N\}$, $i_2 = \min\{i : \xi_i \geq \rho, i_1 + 1 \leq i \leq N+1\}$.*

According to Corollary 1, the optimal solution to the DRO newsvendor problem under the JW ambiguity set $\mathcal{F}_\phi(\hat{F}, \rho)$ can be easily found.

When ϕ is the variational distance (i.e., $\phi(t) = |t - 1|$) and the distribution is discrete, the expression of q^r is the same as that in proposition 7 in Bertsimas et al. (2018). Therefore, Theorem 1 can be viewed as an extension of that result. Theorem 1 provides a general framework that holds for both continuous and discrete reference distributions and a general class of discrepancy measures. In Subsection 5.2, we compare the solution using the KS distance with other choices of discrepancy measures. The result shows that the solution using the variation distance (KS distance) seems extremely conservative, whereas other choices of ϕ functions provide more reasonable solutions. This observation is consistent with the claim in Mason and Schuenemeyer (1983) that alternative choices of the ϕ-divergence provided in the JW ambiguity set can mitigate the disadvantage of the KS distance being more sensitive near the center than at the tails. Moreover, the JW ambiguity set can be perfectly applied to the feature-based newsvendor problem, which is the focus of Section 6.

As mentioned in Section 3.2, any ∞-Wasserstein ball or Lévy ball ambiguity set is an FSD ambiguity set. Directly using Theorem 1, we can also obtain the closed-form inventory solution under the ∞-Wasserstein ambiguity set or Lévy ball ambiguity set. For the sake of brevity, we only present conclusion of ∞-Wasserstein ball here.

Corollary 2. *Assume $\mathcal{F}$ is the ∞-Wasserstein ball of a reference distribution $\hat{F}$ with radius ρ; the optimal solution of the DRO Problem (1) with loss function l_0 is $q^r = \min\{d : \hat{F}(d) \geq \frac{b}{b+h}\} + \frac{b-h}{b+h}\rho$.*

Clearly, the optimal inventory level for the ∞-Wasserstein ambiguity set is the newsvendor inventory level for the reference distribution shifted by an amount $\frac{b-h}{b+h}\rho$, which is proportional to the robustness level ρ.

Another important special case of Theorem 1 is summarized in the following corollary.

Corollary 3. *Assume only the range of demand $[d_l, d_u]$ is known—that is, $\mathcal{F}$ is the set of all possible distributions such that $d_l \leq \tilde{d} \leq d_u$ almost surely, the optimal solution of the DRO Problem (1) with loss function l_0 is $q^r = \frac{bd_u + hd_l}{b+h}$.*

Corollary 3 means that if only the upper and lower bounds of demand are available, the closed-form solution reduces to the optimal solution of the stochastic newsvendor problem when the demand is uniformly distributed between these bounds. This result was first derived by Morris (1959) and then studied by Kasugai and Kasegai (1961) and Vairaktarakis (2000). Corollary 3 provides a new derivation and perspective on this classic result.

The existence of a closed-form optimal inventory decision means that computing DRO optimal inventory decisions under the JW ambiguity set and the ∞-Wasserstein ambiguity set is as easy as solving the classic newsvendor problem. In general, however, it is necessary to solve at least a convex optimization problem for the DRO newsvendor problem under the ambiguity set based on ϕ-divergence (Ben-Tal et al. 2013) or the Wasserstein distance (Gao and Kleywegt 2023) with

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a discrete bounded sample space. In addition, the worst-case distribution of the ϕ-divergence ambiguity set may have spikes (cf. Gao and Kleywegt 2023), as it does not consider the order of the demand data. For the Wasserstein distance, Esfahani and Kuhn (2018) proved that the DRO solution with the 1-Wasserstein ambiguity set with sample space $\mathbb{R}$ coincides with the SAA solution. The resulting worst-case demand distribution might have negative support. Lee et al. (2021) considered the p-Wasserstein ambiguity set ($p \geq 1$) with nonnegative demand support and derived a closed-form solution for the case in which $b \geq h$, but the closed-form solution is not known if this condition is not satisfied. Compared with Scarf's mean-variance ambiguity set, the size of our proposed ambiguity set is adjustable by tuning ρ according to the number of data points. Recently, Besbes and Mouchtaki (2023) investigated the robust newsvendor policy that optimizes the worst-case relative regret. Specifically, they proposed a minimax policy, which is a convex combination of consecutive order statistics. Although our optimal inventory level pertains to a different problem setting, it shares a similar structure with theirs. It is also worth noting that the two order statistics in our optimal closed-form solution are not necessarily consecutive order statistics, highlighting some distinctions between our solution and theirs.

As an extension, we investigate a multidimensional DRO problem where the objective function is the sum of separable monotone or convex functions. The ambiguity set contains all joint distributions with marginal distributions that belong to JW ambiguity sets. A polynomial-size linear program reformation is derived and applied to the distributionally robust vehicle allocation problem. For further details, see Online Appendix F.

4.1. The Perceived Critical Ratio

The closed-form solution allows us to investigate the properties of the DRO solutions for different reference distributions. We choose the ambiguity set to be JW ambiguity set based on χ_2 distance in the remainder of this section. Suppose the reference distribution has CDF $F(d)$ and the critical ratio is η. Let

$$\eta^l = \left(2\eta + \rho - \sqrt{4\eta(1-\eta)\rho + \rho^2}\right)/(2+2\rho);$$
$$\eta^u = \left(2\eta + \rho + \sqrt{4\eta(1-\eta)\rho + \rho^2}\right)/(2+2\rho).$$

We can see from Theorem 1 and Table 2 that the optimal DRO inventory solution is $q^r = \eta F^{-1}(\eta^u) + (1-\eta)F^{-1}(\eta^l)$. This means that the DRO solution is determined by a convex combination of the η^l and η^u quantiles of the reference distribution, and the length of the interval $\eta^u - \eta^l$ increases with respect to ρ. To deliver more insights, we define the perceived critical ratio as $\eta^r \triangleq F(q^r)$, which is the critical ratio that makes the newsvendor quantity under the reference distribution equal q^r. In other words, the optimal DRO inventory solution is the η^r quantile of the reference distribution. Therefore, $\eta^r < \eta$ means that the DRO optimal quantity is smaller than the optimal order quantity based on the reference distribution, whereas $\eta^r > \eta$ means the opposite.

We start with the simplest case, where the reference distribution is the uniform distribution.

Example 2 (Uniform Distribution). Assume that the ambiguity set is $\mathcal{F}_\phi(\hat{F}, \rho)$, where $\hat{F}$ is the uniform distribution on $[a, b]$, where $a \geq 0$ and ϕ is the χ_2 distance; then, the perceived critical ratio is

$$\eta^r = \eta + \frac{\eta - 0.5}{1+\rho}\left(\sqrt{4\rho\eta(1-\eta) + \rho^2}) - \rho\right). \tag{5}$$

From Equation (5), it is clear that when the critical ratio $\eta > 0.5$, which means the underage cost is higher than overage cost, we have $\eta^r > \eta$. Similarly, when $\eta < 0.5$, we have $\eta^r < \eta$. This property implies that the ambiguity-adverse decision maker will adjust the order quantity away from the median.

Definition 8. The DRO inventory solution satisfies the *upper-median-repulsion* (UMR) property if $\eta^r > \eta$ whenever $\eta > 0.5$. Similarly, it satisfies the *lower-median-repulsion* (LMR) property if $\eta^r < \eta$ whenever $\eta < 0.5$. The *median-repulsion* (MR) property holds if both UMR and LMR hold.

In the next proposition, we provide some conditions under which the aforementioned properties hold.

Proposition 4. *Assume the reference distribution is continuous and $\rho > 0$.*

1. *The MR property holds if the probability density function (PDF) of the reference distribution is symmetric and unimodal.*

2. *The UMR property holds if the CDF of the reference distribution $\hat{F}(d)$ is concave in the region*:

$$\hat{F}(d) \in \left[\left(2\eta + \rho - \sqrt{4(1-\eta)\eta\rho + \rho^2}\right)/(2+2\rho),\right.$$
$$\left.\left(2\eta + \rho + \sqrt{4(1-\eta)\rho + \rho^2}\right)/(2+2\rho)\right].$$

3. *The LMR property holds if the CDF of the reference distribution $\hat{F}(d)$ is convex in the region*:

$$\hat{F}(d) \in \left[\left(2\eta + \rho - \sqrt{4(1-\eta)\eta\rho + \rho^2}\right)/(2+2\rho),\right.$$
$$\left.\left(2\eta + \rho + \sqrt{4(1-\eta)\rho + \rho^2}\right)/(2+2\rho)\right].$$

Part (1) of Proposition 4 shows that the MR property holds for any distribution that has a single peak at the middle, such as the normal distribution, t-distribution, and symmetric triangular distribution. Based on parts

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(2) and (3), the convexity or the concavity of the CDF over a specific line segment has an implication on the UMR or LMR. In particular, for distributions such as exponential and Pareto that have a concave CDF, the UMR always holds. On the other hand, for distributions such as the beta distribution with $\alpha \geq 1$, $\beta \leq 1$ that have a convex CDF, the LMR holds.

We conclude this section by providing some additional results for the case where the reference distribution is exponential. It is not hard to verify that both $\kappa^{-l}_{\phi,\rho}(\hat{F})$ and $\kappa^{-u}_{\phi,\rho}(\hat{F})$ are Lebesgue integrable on $[0,\infty)$; thus, the condition in Proposition 1 holds for exponential reference distribution. Similar to the uniform distribution case, we can derive a closed-form expression for the perceived critical ratio.

Example 3 (Exponential Distribution). Assume that the ambiguity set is $\mathcal{F}_\phi(\hat{F},\rho)$, where $\hat{F}$ is the exponential distribution and ϕ is the χ_2 distance; then, the perceived critical ratio is

$$\eta^r = 1 - \frac{1}{2+2\rho}\left(2-2\eta+\rho-\sqrt{4\eta(1-\eta)\rho+\rho^2}\right)^{\eta} \left(2-2\eta+\rho+\sqrt{4\eta(1-\eta)\rho+\rho^2}\right)^{1-\eta}.$$

Because the CDF for the exponential distribution is concave, based on Proposition 4, the UMR holds; that is, whenever $\eta > 0.5$, $\eta^r > \eta$. The following proposition provides some insights for the case where $\eta < 0.5$.

Proposition 5. *Assume that the ambiguity set is $\mathcal{F}_\phi(\hat{F},\rho)$, where $\hat{F}$ is the exponential distribution and ϕ is the χ_2 distance. For any robustness level $\rho > 0$, there exist an $\eta_0(\rho) \in [(1-\rho-\sqrt{\rho(1+\rho)})/2, 0.5]$, such that when $\eta > \eta_0(\rho)$, $\eta^r > \eta$ and when $\eta < \eta_0(\rho)$, $\eta^r < \eta$. Moreover, $\eta_0(\rho)$ decreases from 0.5 to 0 when ρ increases from 0 to $e-1$. If ρ is larger than $e-1$, $\eta_0(\rho)$ becomes 0.*

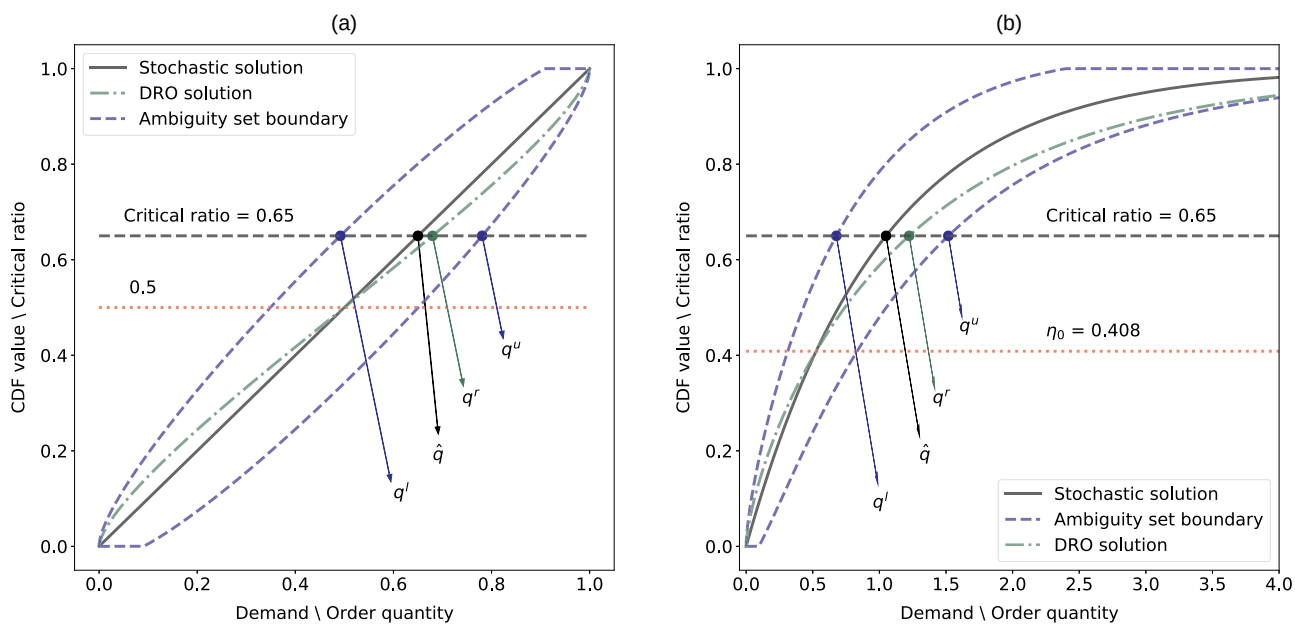
Proposition 5 shows that there exists a threshold η_0, as a function of ρ that the distributionally robust perceived critical ratio is repulsed from η_0. However, η_0 is no longer 0.5 because the one-sided tail of the exponential distribution makes the underage cost play a more significant role in the worst case. Such impact increases when the robustness level ρ increases.

Figure 2 displays the repulsion property of the DRO order quantity q^r versus the optimal order quantity of the reference distribution $\hat{q}$. The y-axis is the CDF value or critical ratio, and the x-axis represents the demand or order quantity. The line represents the CDF, as well as the optimal order quantity under reference distribution. The area between the dashed lines is the distributionally ambiguity set. The dashed dotted line indicates the DRO order quantities. The figure shows examples of order quantities when the critical ratio is 0.65. We can clearly see that the DRO perceived critical ratio is repulsed from 0.5 for the uniform distribution and 0.408 for the exponential distribution.

5. Numerical Study

This section examines the structural insights and performance of the proposed DRO formulation through numerical experiments.

Figure 2. (Color online) DRO and Stochastic Solutions with Different Critical Ratios Under Uniform and Exponential Distribution



Notes. (a) Uniform distribution. (b) Exponential distribution.

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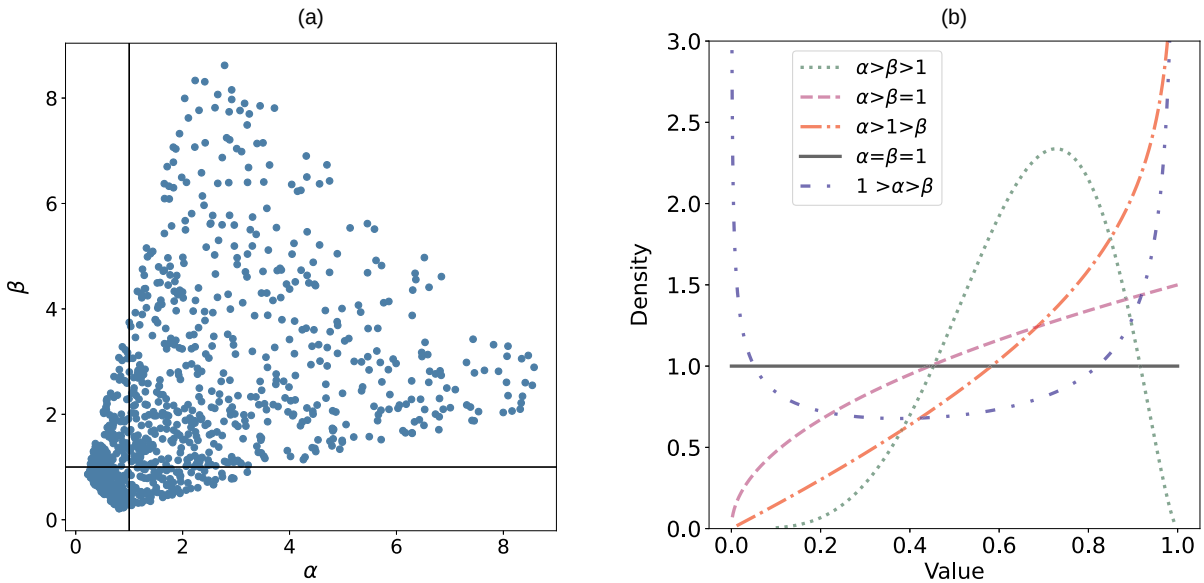
Figure 3. (Color online) Demand Distributions in Numerical Study



Notes. (a) Samples of parameters of beta distribution. (b) PDF of beta distributions with different parameters.

5.1. Performance with Different Data Sizes

To evaluate inventory decisions under different data sizes, we draw demand samples from a ground-truth distribution and vary the number of samples, N, from 5 to 105. We compare the performance of SAA, Scarf's mean-variance approach, and our DRO approach (using the JW ambiguity set based on χ^2-distance). In order to choose a sequence of decreasing robustness levels ρ_N for different data sizes N, we first choose ρ for some specific N from some candidates using cross-validation and then fit the function $\rho_N = (a\ln(N) + b)/N$. This rate is chosen according to the result in Online Appendix E.2. We record the expected loss under the ground-truth distribution and repeat the procedure 1,000 times. At each time, the ground-truth distribution is set to be the beta distribution with random parameters $\tilde{\alpha}$ and $\tilde{\beta}$, which lead to distributions with different PDF shapes. Figure 3(a) shows the 1,000 parameter pairs (α, β) in the sample used in the experiment, and Figure 3(b) displays some examples of the PDFs of the beta distributions with different parameters. For different values of α or β, the PDF shapes of the beta distribution can be bell-shaped, U-shaped, strictly increasing/decreasing, or uniform.

To evaluate the performance of the different methods, we obtain the results of 500 repetitions and consider three different criteria: the mean of the expected loss, the α conditional value at risk (CVaR) of the expected loss, and the mean of the α-CVaR loss after 500 experiments. For each experiment, we standardized the true optimal expected loss as one. Figure 4 shows the results for the cases in which the fixed critical ratios $\frac{b}{b+h}$ are 0.25, 0.5, and 0.75.

As shown in these figures, our DRO approach generally performs better under all three criteria than the SAA solution, especially when the amount of historical data is small. The widely used mean-variance ambiguity set cannot converge to the optimal loss because it only captures partial information about the distribution. In Figure 4, (b) and (e), although the mean-variance ambiguity set performs best when there are fewer than 20 observations, the average expected loss decreases only slightly with additional data. However, in Figure 4(h), the mean-variance ambiguity set yields a surprisingly low CVaR value when the critical ratio is 0.5, possibly because for beta distribution, the mean is always closer to the less dense side than the median (see Payton et al. 1989).

5.2. Decisions with Different ϕ Functions

In this section, we compare the order quantity of our DRO solution with different ϕ functions with a decision using the ϕ-divergence ambiguity set. In the JW ambiguity set, we choose the following ϕ-functions: ϕ^1_{CR}(Kullback-Leibler), $\phi^{-1}_{CR}(\chi^2)$, and variation distance. In the ϕ-divergence ambiguity set, we choose χ^2 distance. The reference distributions are set to be discrete with 200 demand points, and the histograms are shown in Figure 5. As noted above, the closed-form solution of our DRO approach with the variation distance (or, equivalently, the ambiguity set formed by the KS distance) is studied in Bertsimas et al. (2018), but our framework provides more flexible choices regarding the ϕ functions. The results presented in Figure 5 are the gaps between the DRO solution and the true optimal solution under different demand distributions with critical ratios

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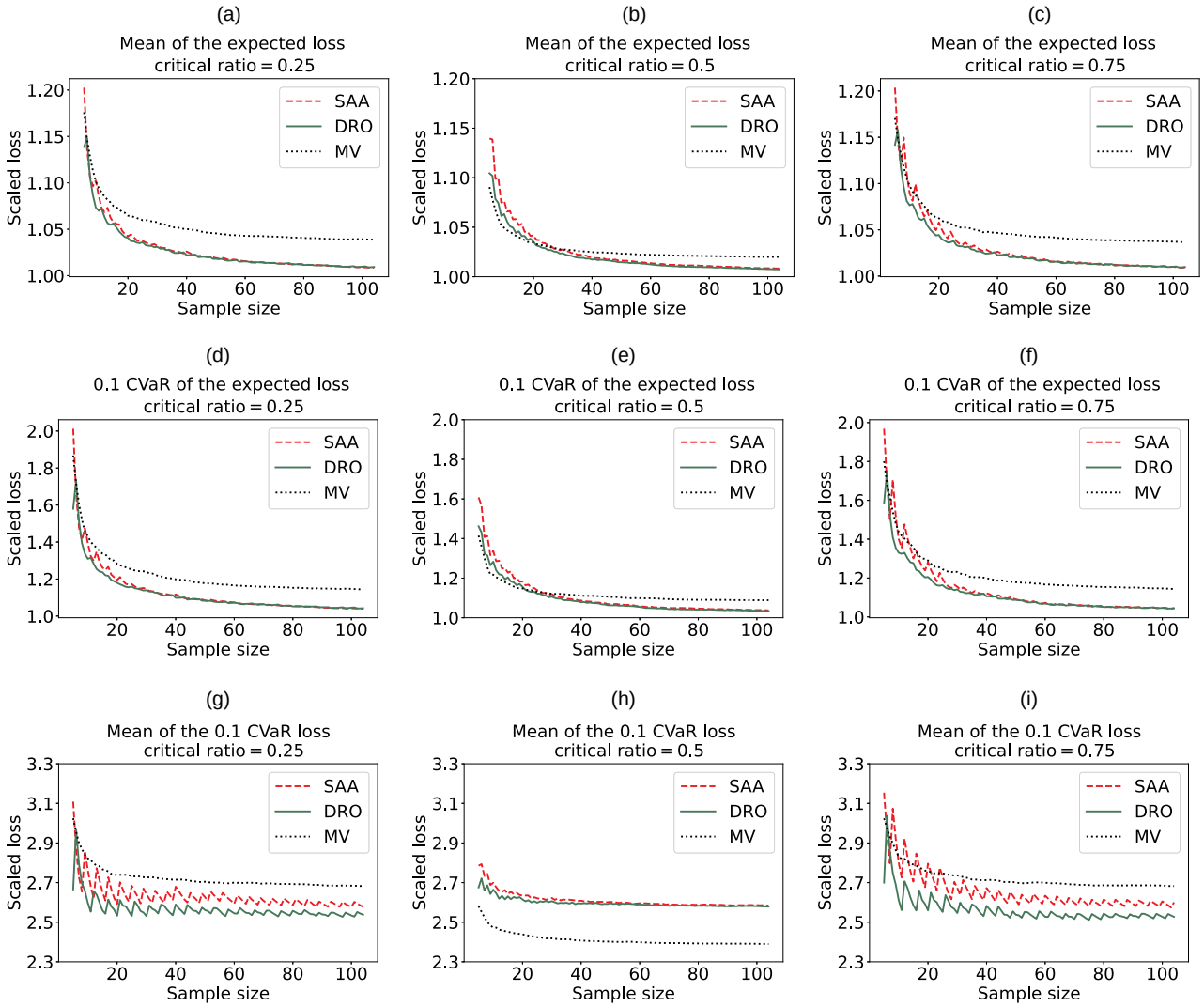
Figure 4. (Color online) Performance of Empirical, DRO, and Mean-Variance Methods Under Different Criteria



Note. Over 500 independent repetitions, for critical ratios of 0.25, 0.5, and 0.75 respectively: (a–c) plot the mean of the expected loss, (d–f) plot the 0.1 CVaR of the expected loss, and (g–i) plot the mean of the 0.1 CVaR loss.

from 0.05 to 0.95. Figure 5 shows that, as expected, the DRO approach tends to order more products when the critical ratio is high and fewer products when the critical ratio is low. However, the solution using variation distance (KS distance) seems to be extremely conservative compared with the other solutions when the critical ratio is lower than 0.2 or higher than 0.8. Note that the parameter ρ is chosen such that the solution behaves less conservatively when the critical ratio is near 0.5. In this sense, the use of ϕ^1_{CR} or ϕ^{-1}_{CR} provides more reasonable solutions. Therefore, the additional flexibility afforded by choosing ϕ in our approach has practical relevance.

The behavior of the ϕ-divergence ambiguity set is similar to other ambiguity sets in case (c). In terms of conservativeness, the ϕ-divergence measure lies between the KS distance and the JW ambiguity set. In case (a), which is analogous to the uniform reference distribution, the ϕ-divergence approach is median-attraction instead of median-repulsion—that is, the DRO order quantity is closer to the median of the distribution than the stochastic order quantity. In addition, there are spikes on the boundary. In case (b), which is analogous to the exponential reference distribution, the order quantity for the ϕ-divergence approach is always larger than that of the stochastic solution. In contrast, other methods order less when the critical ratio is low compared with other scenarios. The reason for these uncommon behaviors for the ϕ-divergence ambiguity set in cases (a) and (b) might be that it does not consider the ordering information of demand points, which leads to spikes on the boundary for the worst-case distribution (Gao and Kleywegt 2023).

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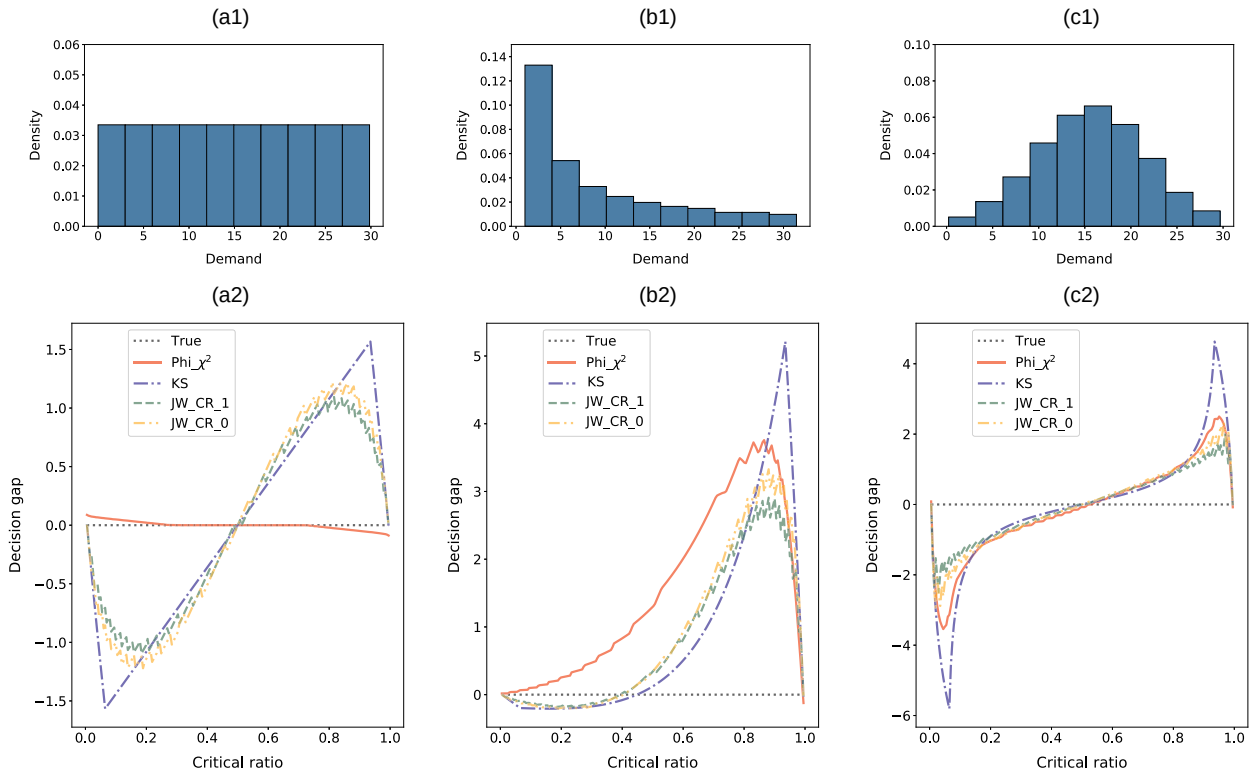
Figure 5. (Color online) Gaps Between the DRO Solution and True Optimal Solution Under Different Demand Distributions



Notes. Plots (a1), (b1), and (c1) show the histograms of discrete reference distributions for three scenarios. Plots (a2), (b2), and (c2) show the gaps between the DRO solution and the true optimal solution with critical ratios ranging from 0.05 to 0.95 for these three scenarios, respectively.

6. Feature-Based Newsvendor with a Case Study

In this section, we apply our developed DRO framework to the feature-based newsvendor problem. Recall that, given the order quantity decision q and the random demand $\tilde{d}$, the newsvendor's loss function is $l_0(q,\tilde{d}) = \max\{b(\tilde{d}-q), h(q-\tilde{d})\}$, where b and h represent the shortage and holding costs, respectively.

In practice, demand usually depends on several observable features, such as weather, location, season, and economic indicators. We assume that the seller has already collected historical data $\{(d_1,\boldsymbol{x}_1),\ldots,(d_n,\boldsymbol{x}_n)\}$, sampled independently from an underlying joint distribution, where d_i and $\boldsymbol{x}_i$ represent the demand and the corresponding feature, respectively, in the i-th observation. We further assume that the feature $\tilde{\boldsymbol{x}}$ from the support set $\mathcal{X}$ is observable before the seller makes the inventory decision.

After observing the current feature $\boldsymbol{x}$, the seller makes an order decision q based on the historical information to minimize the expected loss $\mathbb{E}[l_0(q,\tilde{d})|\boldsymbol{x}]$. Here, the expectation is taken over the random demand $\tilde{d}$ conditioned on $\boldsymbol{x}$. As the exact demand distribution is unknown, we apply the DRO framework, assuming that the real conditional demand distribution lies in an ambiguity set that depends on the demand covariates. In other words, the problem we want to solve is

$$\min_{q} \sup_{F\in\mathcal{F}|\boldsymbol{x}} \mathbb{E}_F[l_0(q,\tilde{d})],$$

where $\mathcal{F}|\boldsymbol{x}$ is the distributional ambiguity set given feature information $\boldsymbol{x}$.

The decision is made in two steps. The first step is to estimate the conditional distribution $F(d|\boldsymbol{x})$ given feature value $\boldsymbol{x}$. This estimator is denoted as $\hat{F}(d|\boldsymbol{x})$. The second step is to adopt our DRO approach based on JW ambiguity set with reference distribution $\hat{F}(d|\boldsymbol{x})$. In the following subsections, we discuss these steps in detail.

6.1. Estimation and Optimization

Assume $(\mathcal{X},\delta)$ is a semimetric space. To obtain an estimator for the conditional distribution of the demand given feature vector $\boldsymbol{x}$, we propose a functional kernel estimation approach based on semimetrics. Kernel estimation is a popular nonparametric estimation method based on weighted nearest neighbors. Formally, the conditional distribution function for the demand derived from our kernel estimation is defined as

$$\hat{F}(d|\boldsymbol{x}) = \frac{\sum_{i=1}^{N} K_w(\boldsymbol{x}_i,\boldsymbol{x})\cdot \mathbf{1}_{d_i\le d}}{\sum_{i=1}^{N} K_w(\boldsymbol{x}_i,\boldsymbol{x})}, \tag{6}$$

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where $\mathbf{1}_{d_i \le d}$ presents the characteristic function and $K_w(\boldsymbol{x}_i, \boldsymbol{x}) = K(\delta(\boldsymbol{x}_i, \boldsymbol{x})/w)$ corresponds to the "closeness" between the new feature $\boldsymbol{x}$ and the historical feature value $\boldsymbol{x}_i$, measured by the kernel function $K(\cdot)$. Here, $\delta(\boldsymbol{x}_i, \boldsymbol{x})$ is the semimetric that measures the distance between $\boldsymbol{x}_i$ and $\boldsymbol{x}$. In this paper, we assume that the kernel function $K(\cdot): \mathbb{R}^+ \longmapsto \mathbb{R}^+$ is defined on $[0,1]$, and, without loss of generality, we assume $\int K(\cdot) = 1$. Some widely used kernel functions include the following: (i) Uniform(box) kernel: $K(u) = \mathbf{1}_{\{u \le 1\}}$; (ii) Triangle kernel: $K(u) = 2(1-u)\mathbf{1}_{\{u \le 1\}}$; and (iii) Quadratic kernel: $K(u) = \frac{3}{2}(1-u^2)\mathbf{1}_{\{u \le 1\}}$.

If $\mathcal{X} \in \mathbb{R}^n$ and the semimetric is defined by the Euclidean norm—that is, $\delta(\boldsymbol{x}_i, \boldsymbol{x}) = \|\boldsymbol{x}_i - \boldsymbol{x}\|$—Formulation (6) is called the Nadaraya-Watson estimator (Nadaraya 1964 and Watson 1964),

$$\hat{F}_{NW}(d|\boldsymbol{x}) = \frac{\sum_{i=1}^{N} K\left(\frac{\|\boldsymbol{x}_i - \boldsymbol{x}\|}{w}\right) \cdot \mathbf{1}_{d_i \le d}}{\sum_{i=1}^{N} K\left(\frac{\|\boldsymbol{x}_i - \boldsymbol{x}\|}{w}\right)}.$$

This estimator is used in Ban and Rudin (2019) to solve the feature-based newsvendor problem. In addition to the classical NW estimator, several machine learning methods, such as the decision tree and the RF, can be used as kernel functions. More details are included in Online Appendix B.

Because the estimated reference distribution is discrete, the distributionally robust optimal solution can be computed with Corollary 1. More specifically, after observing a new feature $\boldsymbol{x}$, the decision maker first computes the distance between the new feature $\boldsymbol{x}$ and each historical feature $\boldsymbol{x}_i$ and then computes ξ_i. At last, the optimal order quantity can be obtained with a simple sorting and searching procedure. In addition, because the robustness level ρ appears only in the last step, our approach is flexible when changing the robustness level. Thanks to the closed-form expression, this procedure of computing the inventory level given feature is as simple as the robust end-to-end policy proposed by Zhang et al. (2024).

Furthermore, we demonstrate that under mild technical conditions, the expected loss for the closed-form optimal solution converges to the actual optimal loss as the size of the data samples increases, with a provable convergence rate. The bounds on the convergence rate provide theoretical guidelines for choosing the robustness level and the bandwidth of the kernels. For the details of this part, please refer to Online Appendix E.

6.2. Case Study

In this section, we apply our method to a real-world data set. The data set comes from JD.com (Shen et al. 2024), a Chinese e-commerce company, and includes order information (price, discount, and quantity), product properties, and customer information for one month. We consider the new product launching problem studied in Lin et al. (2022). We need to make the inventory decision for some products without historical samples. Some information about demand distributions for our focal products can be inferred from the sales of other products based on their features.

To form the product set, 2,000 products are selected based on monthly sales between 5 and 3,500 units. The sales of 1,000 randomly chosen products are used as the training data set and the remaining 1,000 as the test set. Product type and two attributes are taken directly from the data set. Additionally, we obtain product features such as average price, average final price, and average discount. We also average the features of the customers (e.g., gender, marital status, age, education level, purchasing power, and city-level) who purchased each product and use those as product features. In practice, the seller can potentially obtain or approximate this information for their new products. The inventory replenishment cycles are assumed to be one month, and demand for new products could only be estimated based on sales of other products. Because of the lack of inventory data, we simply assume that the observed sales are the true demand without considering the possibility of censored demand. Because the amount of data is limited compared with the feature dimensions, and many of the features are binary variables, the traditional Euclidean distance may not be a good choice in kernel estimation. Instead, we choose the proximity measure derived from the random forest algorithm and triangle kernel—that is, Formulation (8) in Online Appendix A—to estimate the reference distribution.

We compare the performance of our model with other benchmarks in terms of the newsvendor loss. The methods can be categorized into three groups. The first group comprises methods that do not incorporate feature information. This includes the nonrobust (or, equivalently, SAA) method, which minimizes the average historical loss (i.e., the $b/(b+h)$-quantile of the empirical demand distribution) and two DRO approaches that utilize JW ambiguity sets based on the empirical demand distribution. The second group explicitly considers the feature information. Based on a reference distribution estimated with the RF method, we also make decisions with the nonrobust method, which directly minimize the loss under the reference distribution, and DRO approaches with JW ambiguity sets and the ϕ-divergence ambiguity sets proposed in Ben-Tal et al. (2013). All DRO methods with JW and the ϕ-divergence ambiguity sets in these two groups are based on χ^2 divergence. Moreover, we consider the empirical risk minimization (ERM) methods with linear decision rules. We include a standard version of ERM and a regularized variant (ERM_Rgl).

Because the overage and underage costs are not available from data, we specify them in the following way. We first consider the case in which the overage and underage costs are identical across different products.

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Table 3. Performance and Running Time of Different Methods with Different Critical Ratio

	No feature			Random forest					Linear rule	
	NR	JW_0.02	JW_0.05	NR	JW_0.02	JW_0.05	Phi_0.01	Phi_0.05	ERM	ERM_Rgl
Running time	<0.01	<0.01	<0.01	19.048	19.056	19.068	191.568	202.694	12.306	12.935
$\eta = 0.2$										
Mean loss	1.000	1.000	1.001	0.954	0.953	**0.948**	0.952	0.952	1.134	0.988
Variance	1.000	1.001	1.001	0.915	0.924	0.924	0.912	**0.904**	1.016	0.972
0.01-CVaR	1.000	1.000	1.000	0.967	0.970	0.970	0.966	**0.959**	0.977	0.989
0.02-CVaR	1.000	1.000	1.000	0.961	0.966	0.968	0.960	**0.956**	0.991	0.989
0.05-CVaR	1.000	1.000	1.001	0.949	0.953	0.952	0.946	**0.942**	1.028	0.985
0.1-CVaR	1.000	1.000	1.001	0.950	0.953	0.947	0.947	**0.944**	1.055	0.983
$\eta = \frac{1}{3}$										
Mean loss	1.000	0.999	1.000	0.920	0.918	0.918	**0.912**	0.917	1.063	0.965
Variance	1.000	1.002	1.001	0.850	0.856	0.842	0.833	**0.827**	0.857	0.934
0.01-CVaR	1.000	1.000	1.000	0.929	0.935	0.931	0.917	**0.914**	0.899	0.975
0.02-CVaR	1.000	1.000	1.000	0.933	0.937	0.916	0.922	0.917	**0.905**	0.972
0.05-CVaR	1.000	1.001	1.000	0.911	0.908	0.907	0.900	**0.898**	0.935	0.962
0.1-CVaR	1.000	1.001	1.000	0.913	0.912	0.914	**0.904**	0.907	0.973	0.955
$\eta = 0.5$										
Mean loss	1.000	1.000	1.002	0.889	**0.878**	0.882	0.890	0.883	1.029	0.942
Variance	1.000	0.998	0.992	0.807	0.756	**0.744**	0.794	0.757	0.761	0.879
0.01-CVaR	1.000	1.000	0.998	0.924	0.883	**0.866**	0.914	0.884	0.718	0.787
0.02-CVaR	1.000	1.000	0.998	0.913	0.873	**0.858**	0.900	0.862	0.984	0.995
0.05-CVaR	1.000	0.999	0.997	0.880	0.856	**0.852**	0.873	0.864	0.862	0.929
0.1-CVaR	1.000	0.999	0.995	0.885	**0.868**	0.870	0.884	0.873	0.900	0.916
$\eta = \frac{2}{3}$										
Mean loss	1.000	1.005	1.018	0.829	**0.810**	0.826	0.816	0.816	1.008	0.901
Variance	1.000	0.981	0.958	0.660	0.611	0.606	0.634	**0.603**	0.673	0.790
0.01-CVaR	1.000	0.995	0.990	0.828	0.817	0.819	0.823	**0.811**	0.830	0.908
0.02-CVaR	1.000	0.995	0.988	0.801	0.766	0.767	0.784	**0.763**	0.820	0.905
0.05-CVaR	1.000	0.991	0.981	0.796	0.760	0.751	0.775	**0.750**	0.804	0.870
0.1-CVaR	1.000	0.986	0.970	0.816	0.776	**0.768**	0.795	0.775	0.833	0.856
$\eta = 0.8$										
Mean loss	1.000	1.019	1.062	**0.739**	0.787	0.818	0.749	0.814	0.930	0.834
Variance	1.000	0.923	0.851	0.550	0.522	**0.486**	0.526	0.521	0.568	0.684
0.01-CVaR	1.000	0.979	0.958	0.799	0.768	**0.729**	0.775	0.758	0.763	0.855
0.02-CVaR	1.000	0.975	0.950	0.733	0.706	**0.680**	0.712	0.698	0.735	0.842
0.05-CVaR	1.000	0.960	0.919	0.690	0.691	**0.678**	0.682	0.696	0.743	0.798
0.1-CVaR	1.000	0.934	0.866	0.709	0.714	0.706	**0.703**	0.729	0.766	0.783

Note. The boldface type indicates the best-performance entry among each row.

The computational results are summarized in Table 3. The nonrobust methods are labeled as NR, while robust methods are labeled as JW_ρ or Phi_ρ, reflecting the ambiguity sets and robustness level ρ. The robustness levels are selected from a set of candidate values through cross-validation on the training data.

The results indicate that the RF method outperforms both the ERM and the regularized ERM methods. In fact, in most cases, the ERM method exhibits even higher mean loss than the SAA method, likely due to its linearity assumption and lack of feature selection. The regularized ERM performs slightly better than the ERM and no-feature methods due to its feature selection effect. Clearly, the RF method has the advantage of modeling the nonlinear relationship between the features and demand. The performance of the two DRO approaches in terms of the mean, variance, and CVaR is similar. When the robustness level is low, the DRO approaches perform better than the RF methods under different criteria, especially when the critical ratio is high. Increasing the robustness level leads to higher average cost, but lower variance and CVaR, making it suitable for decision makers seeking safer decisions. Moreover, regarding the improvement by DRO approaches, the results show that, without accounting for feature information, the DRO approaches can slightly decrease risk measures and have a negligible effect on reducing mean loss. The efficacy of the DRO approach becomes more pronounced when covariate information is considered. Finally, in terms of running time, thanks to the closed-form solution, our DRO approach takes almost the same amount of time as the nonrobust RF methods. In contrast, much longer time is needed to achieve a similar performance using the DRO approach with the ϕ-divergence ambiguity set. The ERM and regularized ERM methods have extremely short run times

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Table 4. Performance and Running Time of Different Methods with Random Critical Ratio

	No feature			Random forest					Linear rule	
	NR	JW_0.02	JW_0.05	NR	JW_0.02	JW_0.05	Phi_0.01	Phi_0.05	ERM	ERM_Rgl
Running time	**0.426**	0.552	0.539	18.599	18.623	18.619	59.882	62.698	801.892	950.686
Mean loss	1.000	0.999	1.007	0.829	**0.814**	0.821	0.818	0.823	0.988	0.931
Variance	1.000	0.978	0.955	0.678	0.628	**0.610**	0.644	0.626	0.687	0.826
0.01-CVaR	1.000	0.990	0.980	0.867	0.838	**0.821**	0.836	0.824	0.854	0.935
0.02-CVaR	1.000	0.991	0.981	0.807	0.771	**0.757**	0.781	0.769	0.814	0.910
0.05-CVaR	1.000	0.988	0.976	0.791	0.762	**0.753**	0.776	0.767	0.810	0.891
0.1-CVaR	1.000	0.984	0.968	0.805	0.774	**0.772**	0.791	0.783	0.854	0.894

Note. The boldface type indicates the best-performance entry among each row.

because the RF methods and DRO approaches require training an RF, which takes longer than solving a linear program.

Next, we consider a more practical case, in which the critical ratios are different for different products. We randomly designate the underage cost b from 0.2 to 0.8 for each product and let the overage cost $h = 1 - b$. We repeat the aforementioned experiments and obtain the following results.

Table 4 demonstrates that implementing JW_0.02 reduces the mean loss by 1.8% and JW_0.05 reduces 0.02-CVaR by 6.2%, compared with the nonrobust approach. Moreover, the computational times for our DRO approach are nearly identical to the nonrobust method. Note that, in this case, the ERM and the regularized ERM must solve a linear program for each product, and, thus, they have extremely long computational times. However, the RF method and the two DRO approaches train the RF only once, independent of the critical ratio.

In conclusion, our approach presents a clear advantage in terms of both newsvendor cost and computational efficiency. The enhancements achieved through our DRO approaches, compared with nonrobust methods, are evident in terms of the newsvendor cost. Although our method's performance aligns with that of the ϕ-divergence approach, it requires significantly lower computational time. This efficiency renders our method especially advantageous for end-to-end applications, enabling quick decision making as the critical ratio varies, without compromising outcome quality.

In addition, we investigate the discrepancy between our robust approach and the risk-averse approach suggested in Lin et al. (2022). The result shows that the risk-averse method, although performing well when the RF method's profit is low, is too conservative when the potential profit is high. Further details can be found in Online Appendix C.

7. Conclusion

This paper presents a DRO approach to solving the newsvendor problem. The ambiguity set, referred to as the JW ambiguity set, is based on the ϕ-divergence between the CDFs of two parts of the random variable. We establish that this ambiguity set can be expressed through stochastic dominance and derive a closed-form optimal solution to the corresponding newsvendor problem. The closed-form solution also applies to the ∞-Wasserstein ball ambiguity set and the Lévy ball ambiguity set. The numerical experiments with a synthetic data set show that, in most instances, our DRO approach performs better than other state-of-the-art approaches. Additionally, a case study with a real-world data set that incorporates feature information shows that our DRO approach provides solutions that perform well in terms of inventory loss and computational time. We also prove that the optimal solution converges almost completely to the true optimal solution.

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